

TELECOM CUSTOMER CHURN ANALYSIS

End-to-End SQL Analytics Project

Understand Customers. Reduce Churn. Drive Retention.

10,000 Total Customers	2,037 Churned Customers	20.37% Overall Churn Rate	4,285 MB Avg Data Used	4 Telecom Partners
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Portfolio-ready documentation for GitHub

This report explains the dataset, SQL database setup, analytical framework, query phases, data model, dashboard, business questions, outputs and recommended repository structure so a visitor can understand the project without opening every SQL statement.

1. Project Overview

The project analyzes telecom customer behavior, usage patterns, telecom-partner performance and geographic trends to identify customer-churn patterns and support data-driven retention strategies. The SQL work is organized around Customer Churn, Telecom Partner, Geographic Analysis, Customer Usage Behavior and Retention Strategy perspectives.

The project is designed for telecom-company management and customer-retention teams, with SQL Server as the database/analysis platform.

Source basis: the supplied SQL script explicitly defines the project as “TELECOM CUSTOMER CHURN & RETENTION INTELLIGENCE” and lists the intended perspectives, end consumer and dataset columns.

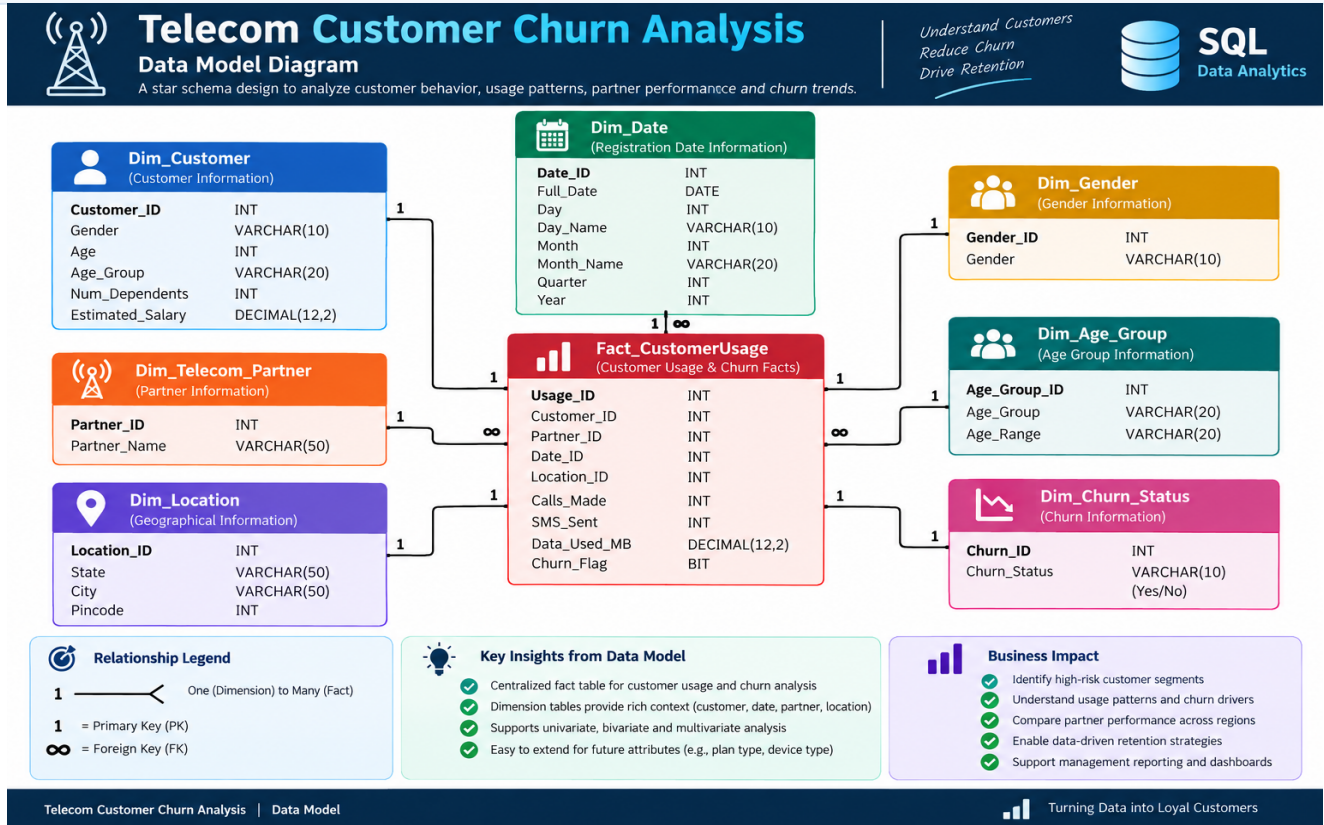
2. Dataset & Database

The core analytical table is **dbo.telecom_churn**. The database setup script creates the SQL Server database **Telecom_Churn_DB**, loads the CSV through a staging table, converts fields to appropriate SQL data types, and then validates the loaded data.

Column	Role / Meaning
customer_id	Customer identifier / primary key
telecom_partner	Telecom partner
gender	Customer gender
age	Customer age
state	Customer state
city	Customer city
pincode	Customer pincode
date_of_registration	Customer registration date
num_dependents	Number of dependents
estimated_salary	Estimated salary
calls_made	Call usage
sms_sent	SMS usage
data_used	Data usage
churn	Churn indicator

3. Data Model & Architecture

The project is analytically centered on a single customer-level table. The generated data-model visual uses a dimensional/star-schema interpretation to explain how customer, date, partner, location, gender, age-group and churn context can organize the analytical concepts. The actual supplied SQL database script, however, creates one main table named `dbo.telecom_churn`; the dimensional tables in the visual are an explanatory analytical model rather than additional tables created by the supplied database script.



4. Analytical Framework

The SQL analysis is divided into four phases, with 15 analyses in each phase. This produces a structured progression from high-level KPIs to one-variable distributions, relationships between two variables, and deeper multi-variable segmentation.

PHASE 1 KPI ANALYSIS High-level customer, churn, partner, geographic and usage KPIs.	PHASE 2 UNIVARIATE ANALYSIS Individual-variable distributions and summaries.	PHASE 3 BIVARIATE ANALYSIS Relationships between churn, usage, partner and geography variables.	PHASE 4 MULTIVARIATE ANALYSIS Combined partner, demographic, usage, geography and retention views.
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Phase 1 — KPI Analysis

- Total customers and total churned customers.
- Overall churn rate.
- Customers by telecom partner and partner churn rate.
- Largest partner by customer base.
- Customer distribution and churn by state/city.
- Average data, calls and SMS usage.
- Average customer age and average number of dependents.

Phase 2 — Univariate Analysis

- Gender, age, dependents, salary and usage distributions.
- Age-group and salary-group segmentation.
- Registration-date distribution.
- Partner, state, city and pincode distributions.

Phase 3 — Bivariate Analysis

- Churn vs gender, age group, dependents and salary group.
- Churn vs calls, SMS and data usage.
- Partner vs customer count, churn and data usage.
- State/city vs churn.
- Gender vs usage and age group vs usage.

Phase 4 — Multivariate Analysis

- Partner + age group + churn.
- Partner + gender + churn.
- Partner + salary group + churn.
- Partner + usage + churn.
- Geography + demographic + usage + churn combinations.

- Retention-priority segmentation and deeper customer profiling.

5. SQL Techniques Used

The supplied SQL script demonstrates practical T-SQL patterns for analytical work, including aggregation, GROUP BY, CASE expressions, CTEs and multi-dimensional grouping. Examples include calculating churn rates with conditional aggregation and creating age/salary groups using CASE.

Representative SQL pattern — partner churn rate

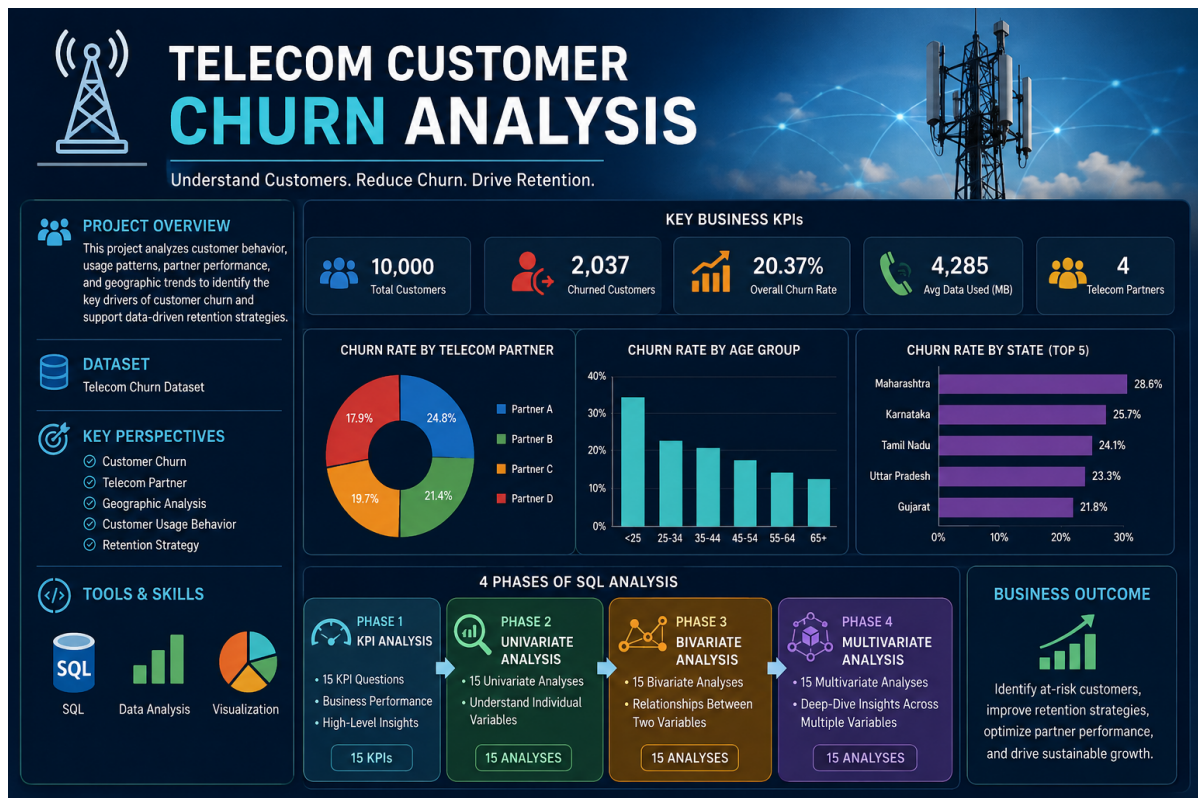
```
WITH T1 AS (  
  SELECT  
    telecom_partner,  
    COUNT(*) AS Total_Customers,  
    SUM(CASE WHEN churn = 1 THEN 1 ELSE 0 END) AS Churned_Customers  
  FROM telecom_churn  
  GROUP BY telecom_partner  
)  
SELECT *,  
  Churned_Customers * 100.0 / Total_Customers AS Churn_Rate  
FROM T1  
ORDER BY Churn_Rate DESC;
```

Analytical strengths

- Conditional aggregation converts the churn flag into counts and rates.
- CASE expressions create reusable analytical groups such as age groups and salary groups.
- CTEs make multi-step analytical calculations easier to read and extend.
- ORDER BY ranking turns raw aggregates into actionable comparisons.

6. Dashboard & Executive View

The supplied dashboard is a presentation layer for the SQL analysis. Its displayed KPI cards and charts are useful for communicating the project quickly to a GitHub visitor or business stakeholder.



Dashboard-displayed values: 10,000 total customers; 2,037 churned customers; 20.37% overall churn rate; 4,285 MB average data used; and 4 telecom partners.

The dashboard also presents churn-rate comparisons by telecom partner, age group and state, followed by the four-phase SQL analysis pipeline and a business-outcome panel.

These KPI numbers are reported as displayed in the supplied dashboard visual; they are not independently revalidated here against the full source dataset.

7. Business Questions & Decision Support

The project is structured to answer practical questions that a telecom management or retention team can use when evaluating customer churn.

Business area	Questions answered
Customer churn	How many customers churned? What is the overall churn rate? Which customer groups show higher churn?
Telecom partner	Which partner has the largest customer base? Which partner has the highest churn rate? How does usage vary by partner?
Geography	Which states and cities have the highest customer concentration or churn rates?
Customer usage	How do calls, SMS and data usage differ across churn status, gender and age groups?
Retention strategy	Which combinations of partner, demographic, salary, usage and geography can be prioritized for deeper retention analysis?

Expected business value

- Identify high-risk customer segments for retention attention.
- Compare telecom-partner performance.
- Understand usage patterns associated with churn.
- Highlight geographic segments requiring closer monitoring.
- Support management reporting and retention-strategy discussions.

8. GitHub Deliverables & Project Workflow

A clean repository should make the project reproducible: a visitor should be able to move from raw data to database setup, SQL analysis, CSV outputs, dashboard and report.

1	Raw CSV	telecom_churn(2).csv
2	Database	Telecom_Customer_Churn_Database.sql
3	Analysis	SQL_QUERY.sql
4	Results	60 query-result CSV files across 4 phases
5	Dashboard	Telecom Churn Analysis Dashboard.png
6	Report	Telecom_Customer_Churn_SQL_Project_Report.pdf

Recommended repository structure

```
Telecom-Customer-Churn-Analysis/  
■  
■■■ database/  
■ ■■■ Telecom_Customer_Churn_Database.sql  
■  
■■■ dataset/  
■ ■■■ telecom_churn(2).csv  
■  
■■■ queries/  
■ ■■■ SQL_QUERY.sql  
■  
■■■ results/  
■ ■■■ Phase_1_KPI/  
■ ■■■ Phase_2_Univariate/  
■ ■■■ Phase_3_Bivariate/  
■ ■■■ Phase_4_Multivariate/  
■  
■■■ dashboard/  
■ ■■■ Telecom Churn Analysis Dashboard.png  
■  
■■■ reports/  
■ ■■■ Telecom_Customer_Churn_SQL_Project_Report.pdf  
■  
■■■ README.md
```

9. How to Run the Project

Step 1 — Prepare SQL Server

- Open SQL Server Management Studio (SSMS).
- Open Telecom_Customer_Churn_Database.sql.
- Change the CSV path variable in the script to the full local path of your telecom_churn(2).csv file.
- Run the database setup script.

Step 2 — Verify the table

- Confirm that database Telecom_Churn_DB exists.
- Confirm that dbo.telecom_churn contains the expected columns and loaded records.
- Use the validation queries included in the database script.

Step 3 — Run the analytics

- Open SQL_QUERY.sql.
- Run the Phase 1, Phase 2, Phase 3 and Phase 4 queries.
- Export each query result to CSV for the GitHub Results folders.

Step 4 — Present the results

- Use the dashboard for executive-level visual storytelling.
- Use this report and the README to explain the methodology and repository structure.

Reproducibility note

The supplied database script uses a staging table to import CSV text and then converts values into typed columns. It also creates a primary key on customer_id and adds analytical indexes for telecom_partner, state and churn.

10. Conclusion

This project demonstrates an end-to-end SQL analytics workflow for telecom customer churn: raw customer data is loaded into SQL Server, organized into an analytical table, transformed through structured SQL questions, exported as query-result CSVs and communicated through a dashboard and portfolio documentation.

The strongest part of the project is its four-stage analytical progression: **KPIs** → **Univariate Analysis** → **Bivariate Analysis** → **Multivariate Analysis**. This structure makes it easy for a reviewer to see how the analysis moves from basic measurement to deeper segmentation and retention-oriented insights.

The supplied SQL file explicitly frames the work around customer churn, telecom partner, geographic analysis, customer usage behavior and retention strategy. The report therefore keeps those perspectives at the center rather than introducing unrelated analytics.

RAW DATA	SQL DAT ABASE	4-PHASE ANALYSIS	CSV RESULTS	DASHBOA RD
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The result is a complete GitHub portfolio project that shows database design, T-SQL querying, analytical reasoning, result generation and business-oriented data storytelling.